

Foundations of Computer Science

CSC-220

About Me

Bailey Cross

Siena/RPI

Quantum Computer/Computer Science
Education



Office: Whitney 239

Office Hours: Tuesday/Thursday 9:30-11:15, Available by appointment

Class Days/Hours: Monday/Wednesday 1:30pm - 2:55pm

2 Exams, Midterm and Final.

Quizzes every module, postings will be on our Brightspace page

Mondays will be lecture with some in-class activities

Wednesdays will be shorter lecture with a lab.

All homework will be due 1 week after it was first assigned.

For example, labs on Wednesdays will be due the next Wednesday at 11:59pm.

What is expected from you?

- On-time
- Attentive
- Participate

Catalog Description

This course focuses on the specification, design, and analysis of software development. This course includes propositional and predicate logic, sets, and relations in how they relate to software development. Students will examine the use of functions, recursive definition and induction, finite state machines, formal languages, Turing machines, and the limits of computability in the design of a software system. This course covers the following topics:

- Logic: Convert logical statements from informal language to propositional and predicate logic expressions; apply formal logic to model and analyze the correctness of software constructions; inductively prove properties of recursively defined functions.
- Discrete structures: Use sets, functions, and relations to model software problems and solutions.
- Formal languages: Design a finite state machine to accept a specified language; transform a non-deterministic machine to a deterministic one; design a regular expression to represent a specified language; recognize and use context-free grammar that represents a small expression language.
- Computability: Recognize uncomputable problems that have no algorithmic solution.

Course Materials

- Critchlow, Carol, and David Eck. Foundations of Computation. (available online)
- Supplementary resources:
 - Lehman, Eric, F Thomson Leighton, and Albert R Meyer. Mathematics for Computer Science. (available online)
 - Michael Sipser, Introduction to the Theory of Computation, Third Edition, 2013, Cengage Learning. ISBN-13: 978-1-133-18779-0.
 - Stein, Cliff L, Robert Drysdale and Kenneth Bogart. Discrete Mathematics for Computer Scientists, Addison-Wesley, 2010.
 - Thomas H. Cormen and Charles E. Leiserson. Introduction to Algorithms, 4th Edition
- A computer or laptop is required for this course. Windows is the preferred operating system

Late Policy

$$\text{Late Grade} = \text{PointsEarned} * (1 - \text{DaysLate}/35)$$

If a student with a 100 turns in an assignment:

1 day late = 97.14

3 days late = 91.43

If a student with an 80 turns in an assignment:

1 day late = 77.71

3 days late = 73.14

After 2 weeks, the assignment gets a 0

Late Factor	Day Percentage	Day	Earned Score	Late Score For 100	Difference	Late Score for 80	Late Score for 50
35	0.971	1	100	97.14	2.86	77.71	48.57
	0.943	2	95	94.29	5.71	75.43	47.14
	0.914	3	90	91.43	8.57	73.14	45.71
	0.886	4	85	88.57	11.43	70.86	44.29
	0.857	5	80	85.71	14.29	68.57	42.86
	0.829	6	75	82.86	17.14	66.29	41.43
	0.800	7	70	80.00	20.00	64.00	40.00
	0.771	8	65	77.14	22.86	61.71	38.57
	0.743	9	60	74.29	25.71	59.43	37.14
	0.714	10	55	71.43	28.57	57.14	35.71
	0.686	11	50	68.57	31.43	54.86	34.29
	0.657	12	45	65.71	34.29	52.57	32.86
	0.629	13	40	62.86	37.14	50.29	31.43
	0.600	14	35	60.00	40.00	48.00	30.00

Accommodations

- The Office of Accessibility Resources (OAR) assists students with documented disabilities of all types. If you have documented accommodations or feel that you need additional time, or a private setting to take a test, please reach out to the OAR office in Coulter Hall (C104) for additional help.
- If English is not your first language, and you are looking for additional support with strengthening everyday English skills, the English Language Institute (ELI) can support you with additional training to build your confidence in reading, writing, listening, and speaking English. In addition, if you are an ELI student, you are eligible for extended time on exams in Testing Services.

- All students are encouraged to use the Learning Center for free academic support. Services include tutoring, study support and academic skill development. They provide drop-in tutoring, scheduled appointments, remote tutoring, and more. The Learning Center is in G202

Cheating/Academic Dishonesty

- This class is governed by the academic rules, which state: "Current and prospective students at Onondaga Community College are expected to adhere to the values of intellectual and academic honesty and integrity. Violations of academic honesty will not be tolerated." Please refer to the Academic Rules in the College Catalog to make sure you know your rights and expectations as a student.

- Any form of cheating will result in an automatic 0 for the assignment for all parties involved. A second offense will result in an automatic F for the course. Students are expected to complete their own assignments. Copying or using data files or code from current or past students, or from any other person or source (including online help services), will be considered cheating. Students should read and familiarize themselves with the BCC Academic Honesty policy

Administrative X

- Any student that stops actively participating in or attending class must be administratively withdrawn from the class with a grade of X. An administrative X grade has negative impacts on financial aid, veterans' benefits, and athletic eligibility and can lead to the student being forced to leave on-campus housing.
- After a student has missed 5 classes consecutively they will be administratively withdrawn from CSC-220.

Use of AI

Red Light: Generative AI Use Prohibited

The use of generative AI, technology that generates new content/outputs in response to a user prompt or request, is not permitted in this course. While generative AI has capabilities, it does not replace the unique value of human critical thinking and problem-solving skills. The development of essential professional skills, such as the ability to produce original ideas, generate hypotheses, and analyze concepts, is central to this course. To foster the development of critical thinking and independent learning, all assignments must reflect the student's original effort, including their original analysis, interpretation, and composition. Presenting AI-generated content as one's own violates academic integrity. Refer to the Academic Rules for policies related to Academic integrity.

Examples of what's not allowed:

Using AI to generate, create, or assist any part of assignments, quizzes, and/or exams.

Using AI to complete problem sets or analyze data.

Using AI to summarize course readings.

Yellow Light: Generative AI Use Permitted in Limited and Specific Contexts

Generative AI, technology that generates new content/outputs in response to a user prompt or request, can be used in this course, but only for specific tasks and with instructor permission. For instance, generative AI might be permitted for brainstorming or editing, but not for drafting entire papers according to assignment instructions and guidelines. Students are encouraged to ask questions about appropriate use. When generative AI is permitted, students must be transparent and follow citation standards of the course. Presenting AI-generated content as one's own violates academic integrity, and all course work must reflect the student's original analysis, interpretation, and composition. Refer to the Academic Rules for policies related to Academic integrity. Students should verify the accuracy, potential bias, and overall reliability of generated content and engage with course material and resources to ensure ethical and responsible use of generative AI when allowed. Additionally, the librarians at Coulter Library are available to help students learn how to assess information, including AI-generated content, and assist with citations. Students remain responsible for ensuring the completeness of their submitted work reflects their own intellectual contributions and understanding of course learning outcomes.

Examples of what's not allowed:

- Generating full drafts of papers or other assignments.
- Using AI to solve math problems or conduct data analysis.
- Completing quizzes or exams with AI.

Examples of allowed use:

- Using AI to brainstorm initial research questions or explore different perspectives on a topic.
- Assistance with grammar, style, and clarity of writing.
- Creating outlines or initial drafts (with substantial revision and development).

Green Light: Generative AI Use Permitted and Encouraged

In this course, students are welcome to use generative AI, technology that generates new content/outputs in response to a user prompt or request. Students will be taught how to utilize GenAI responsibly and ethically to support learning and exploration of course concepts. Students are expected to develop AI literacy, which includes understanding how AI tools work, evaluating their output critically, and recognizing potential biases. To prepare students for success in an increasingly AI-driven workforce, this course emphasizes the use of AI tools responsibly and productively. When generative AI is used, students must be transparent and follow citation standards of the course. Presenting AI-generated content as one's own violates academic integrity. Refer to the Academic Rules for policies related to Academic integrity. All assignments must reflect the student's original analysis, interpretation, and composition. Students are responsible for verifying the accuracy, potential bias, and reliability of generated content and for using generative AI responsibly. Additionally, the librarians at Coulter Library are available to help students learn how to assess information, including AI-generated content, and assist with citations. Generative AI can support and enhance student engagement in this course.

Examples of what's not allowed:

- Submitting AI-generated work without proper citation.
- Using AI to misrepresent data.
- Violating copyright or intellectual property rights with AI-generated content.

Examples of allowed use:

- Brainstorming ideas and developing outlines.
- Analyzing data and creating visualizations.
- Developing study aids and practice questions.

How to write a professional email

Subject Line is clear and has uppercase for all words **except** for articles, prepositions, and conjunctions

Articles: the, a , an

Prepositions: to, of, in ,on ,for

Conjunctions: and, but or, so

- Body starts out with a greeting "Dear Prof.", "Good Morning/Afternoon Dr."
- Followed by a comma
- Be quick and to the point. Context is sometimes necessary, but too much will seem like oversharing
- End the email with an address of their time and thank them for their consideration

Example

Dear Prof. Cross,

My miniature horse service animal just had surgery and I have to pick them up during class time.

I just want to schedule a time outside of your office hours to talk about the material for this week. Thank you for your time and consideration of my request.

Best,

Bailey

Group activity

Get together in groups of 4 or 5 and talk about your hobbies and passions

Take the pre-class survey

Create a quick presentation on powerpoint or google slides (no more than 4 minutes)

Next class we will present them, no professional attire

Assignments/Homework

Pre-Class Survey